Assignment Topic : Co-ordinate

Introduction to Coordinates

Coordinate Geometry is an important branch of mathematics where geometric shapes such as points, lines, triangles, and circles are represented using numbers. This system allows us to identify the exact position of any given point easily. The coordinate system was introduced by **René Descartes**, and hence, it is known as the **Cartesian Coordinate System**.

Cartesian Coordinate System

In the Cartesian coordinate system, there are two perpendicular axes:

- **X-axis:** The horizontal axis.
- **Y-axis:** The vertical axis.

The point where these two axes intersect is called the **Origin**, denoted as (0, 0).

What is a Coordinate?

A coordinate is an ordered pair (x, y) that represents the position of a specific point.

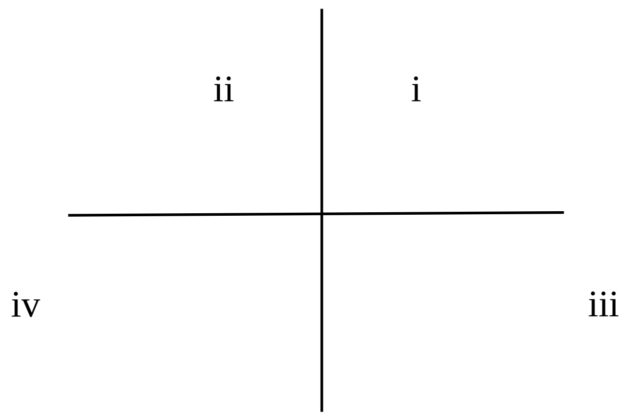
- **x-coordinate:** Indicates how far the point is from the X-axis. • **y-coordinate:** Indicates how far the point is from the Y-axis.

For example, the point **P(3, 4)** means:

- It is **3 units** to the right of the X-axis (if positive) or left (if negative).
- It is **4 units** above the Y-axis (if positive) or below (if negative).

Quadrants

The Cartesian plane is divided into four quadrants:



I (First Quadrant): (x,y)=(+,+)
II (Second Quadrant): (x,y)=(−,+)
III (Third Quadrant): (x,y)=(−,−)
IV (Fourth Quadrant): (x,y)=(+,−)

Coordinate Geometry Formulas

The formulas of coordinate geometry help in conveniently proving the various properties of lines and figures represented in the coordinate axes. The formulas of coordinate geometry are the distance formula, slope formula, midpoint formula, section formula, and the equation of a line.

Applications of Coordinate Geometry

1. Finding the distance between two points
2. Finding the midpoint of a line segment
3. Determining the slope of a line
4. Calculating the area of a triangle
5. Checking the collinearity of points

Distance Formula

The distance between two points $(x_1,y_1)(x1,y1)$ and $x_2,y_2)x2,y2)$ is equal to the square root of the sum of the squares of the difference of the x coordinates and the y-coordinates of the two given points. The formula for the distance between two points is as follows.

$$D = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

Mid-Point Formula:

The formula to find the midpoint of the line joining the point (x_{1}, y_{2}) and (y_{2}, y_{2}) is a new point, whose abscissa is the average of the x values of the two given point, and the ordinate is the average of the y values of the two given points. The mid point lies on the line joining thatwo points and is located exactly between the two points,

(x, y) = ((x_{1} + x_{2})/2, (y_{1} + y_{2})/2)

Area of a Triangle Coordinate Geometry Formula

The area of a triangle having the vertices A$(x_1,y_1)(x1,y1)$, B$(x_2,y_2)(x_2,y_2)$, and C$(x_3,y_3)(x_3,y_3)$ is obtained from the following formula. This formula to find the area of a triangle can be used for all types of triangles.

Area of a Triangle = $\frac{1}{2}(x_1 y_2 + x_2 y_3 + x_3 y_1 - x_1 y_3 - x_1 y_3 - x_3 y_2 - x_2 y_1)$

Problem :

Fop what value of k are the points (2,3), (-9,-6) and (K , 12) Collinear.

Solution:

We know,

$\triangle = 0$

$$\begin{pmatrix} 2 & 3 \\ -4 & -6 \\ k & 12 \\ 2 & 3 \end{pmatrix}$$

Let,

$\triangle = \frac{1}{2}$ (-12-48+3k-24+6k+12)

$= \triangle \frac{1}{2}$ (9k - 72)

$\therefore 0 = 9k - 72$

$\therefore 9k = 72$

$\therefore k = 8$

Section Ratio Formula:

The co-ordinate of a. point R(x, y) which divides the join of two point p(x,y) and Q(x_2,y_2) in a given ratio m_1: m_2.

Internally formula

$$X=\frac{m_1x_2+m_2x_1}{m_1-m_2} \quad y = \frac{m_1y_2+m_2y_1}{m_1-m_2}$$

References

- Coordinates. (2022). Retrieved from cuemath: https://www.cuemath.com/
- Rahman, A. F., & Bhattacharjee, P. (1975). In *co-ordinate geometry (Two And Three Dimensions).* Chittagong : S. Bhattacharjee, .